

Project 13

Repeated Contour Execution on custom_3R

Format: Group of two students; simulation-based project.

Course foundations: Lectures 3, 5–7; Exercise 3.

Motivation

A closed workspace path need not return a cuspidal robot to its starting posture. The next traversal then begins from a different IK and may behave differently. Repeatability is therefore a property of the lifted joint motion, not simply the visible closure of a Cartesian contour.

Task

Overall objectives. Track repeated position contours on the exact custom_3R robot and explain joint-space closure, posture changes, and failed continuation using reduced singularity maps and the workspace IK distribution.

- **Prepare the model and maps.** Validate the exact orthogonal, non-intersecting-axis custom_3R kinematics. Plot $\det J_p$ and its zeros in (q_2, q_3) , map the corresponding critical curves into (ρ, z) , and identify the two-/four-IK workspace regions with representative IK overlays.
- **Choose two closed position curves.** Use an instructor-supplied curve exhibiting nontrivial continuation and one nearby comparison curve. Alternatively construct candidates with the Lecture 6 exact-model lab. Verify closure in three-dimensional Cartesian coordinates; the Lecture 7 idealized square must be revalidated before transferring it to this robot.
- **Lift successive laps.** Enumerate the starting IKs and continue each admissible one for up to three laps. Start every new lap at the actual configuration reached by the previous lap, without resetting the seed. Track solutions by continuity and stop or refine when a regular continuation can no longer be recovered.
- **Classify the observed outcomes.** Distinguish joint-space closure, return to a different IK of the same target, and termination. At each lap boundary, match the reached configuration to the starting-pose IK set modulo joint periodicity, and display the observed mapping. Do not assume every curve defines a permutation if some lifts terminate.
- **Explain events globally.** Overlay all lifted paths on the reduced singularity map and relate events to crossings of workspace critical curves and changes in IK count. A critical-value crossing does not force every possible lift to be singular; evaluate the Jacobian of the selected lift. Refine the decisive intervals.

Specifics and scope

Use two loops, up to four initial mathematical arm IKs, and at most three laps per initial solution. Separate joint-limit feasibility from unconstrained topology. Reuse the analytical IK and continuation tools; discovering a new cusp or proving a general repeatability theorem is outside scope. A predicted outcome that fails numerical validation must be reported rather than imposed.

Use the position Jacobian J_p for the 3R task. Confirm that the meridian reduction is valid for the selected model and operational point. Count distinct real IKs modulo periodicity, then filter joint limits separately. Identify two-/four-IK regions where present and explain any absent count rather than forcing both types into every map. Show both determinant values and the zero contour, and distinguish sampled connectivity from a proof. Refine the decisive region instead of claiming completeness from a coarse grid.

Expected outcome

Understand why repeated Cartesian motions can have different joint-space outcomes. Deliver lap-by-lap configuration records, IK endpoint associations, determinant and workspace-count plots, and an explanation of every observed closure, posture change, or termination.

Submit reproducible code, model parameters and test data, the required plots, a concise 5–6-page report, and a short simulation demonstration. Explain both successful cases and failures; conclusions must follow the numerical and geometric evidence rather than an expected outcome.

Resources

The exact custom_3R is an orthogonal 3R robot with non-intersecting consecutive joint axes. Its URDF is `lectures_main/assets/models/custom_3R/custom_3R_new.urdf`; STL meshes are in the same directory. Use the lecture website to inspect axes and frames, verify D–H parameters, plot IKs, and animate configurations. Lecture 7 provides the repeated-loop concept, while Lecture 6 contains an exact custom_3R motion example. Two suitable starting curves or seed data will be supplied if the existing demonstration cannot be exported directly.

Use the lecture website to verify D–H parameters, visualize IKs and singularities, and simulate paths. All listed paths are relative to the repository root. Start the website following `lectures_main/README.md`.

Workload and collaboration

Budget approximately 60 hours per student (120 person-hours per pair) for this 2-ECTS project, following EPFL’s 30-hours-per-credit convention.¹ Deduct any lectures or exercises included in those credits. Allocate roughly 20 team-hours to model validation, 50 to the core work, 30 to experiments, and 20 to reporting. Reuse the specified IK and simulations. Share validation and interpretation even if modelling and implementation are divided between students. Hardware execution is outside scope.

¹EPFL: Design a Course, workload guidance.